

Reanalysis of Scheibehenne, Jamil, & Wagenmakers (2016)

Hotel towel reuse

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Overview

- Computes joint BFs for the set of experiments reported in from Scheibehenne et al. (2016).
- Source paper: Scheibehenne, B., Jamil, T., & Wagenmakers, E.-J. (2016). Bayesian Evidence Synthesis Can Reconcile Seemingly Inconsistent Results: The Case of Hotel Towel Reuse. *Psychological Science*, 27(7), 1043-1046.
- Each experiment is analyzed as one 2x2 contingency table of counts:
 - rows: condition (control, social norm)
 - columns: towel reuse outcome (yes, no)
- For each study, evidence is computed for:
 - predicted hypothesis: social norm > control (higher reuse probability)
 - unconstrained hypothesis
 - complement: social norm < control

```

library(BayesFactor)
library(dplyr)
library(here)

source(here("functions/compute_joint_complements.r"))

# helper functions for formatting numbers
fmt2 <- function(x) sprintf("%.2f", x)
fmt4 <- function(x) sprintf("%.4f", x)
fmtE <- function(x) sprintf("%.2e", x)

```

Data

```

rawData <- read.table(
  here("data/towelData.csv"),
  header = TRUE,
  sep = ";",
  stringsAsFactors = FALSE
)

# n experiments, each represented by 4 rows (from 2x2 contingency table)
n <- nrow(rawData) / 4

# Add pooled "Total" 2x2 table as in the original script
bigTable <- array(rawData$Count, dim = c(2, 2, n))
total2x2 <- apply(bigTable, 2, rowSums)

totalFrame <- data.frame(
  Source = rep("Total", 4),
  AuthorName = rep("Total", 4),
  Experiment = rep(NA, 4),
  Year = rep(NA, 4),
  Group = c("Control", "Social Norm", "Control", "Social Norm"),
  Towel.Reuse = c("Yes", "Yes", "No", "No"),
  Count = c(total2x2[1, 1], total2x2[2, 1], total2x2[1, 2], total2x2[2, 2])
)

rawData <- rbind(rawData, totalFrame)

```

Study-specific Bayes factors

```

# Names of experiments
exList <- unique(rawData$Source)

# Initialize vectors to store study-specific Bayes factors
bf_1u_vec <- rep(NA_real_, length(exList))
bf_cu_vec <- rep(NA_real_, length(exList))
bf_10_vec <- rep(NA_real_, length(exList))

```

```

bf_1c_vec <- rep(NA_real_, length(exList))

# Set seed for reproducibility
set.seed(45)

# Compute BFs for each experiment
for (i in seq_along(exList)) {

  # Extract the 2x2 contingency table for the current experiment:
  # rows = condition (control, social norm), columns = towel reuse (yes, no)
  thisEx <- rawData[rawData$Source == exList[i], ]
  table2x2 <- cbind(yes = thisEx[1:2, "Count"], no = thisEx[3:4, "Count"])
  rownames(table2x2) <- c("control", "socialNorm")

  # Fit the unconstrained model  $H_u$  against the null model  $H_0$ :
  # - sampleType = "indepMulti", fixedMargin = "rows": the number of guests per
  #   condition is fixed, so each condition is an independent binomial with its
  #   own reuse proportion ( $\theta_c$  for control,  $\theta_e$  for social norm)
  # - priorConcentration = 1: independent uniform Beta(1, 1) priors on  $\theta_c$ 
  #   and  $\theta_e$  (as in Scheibehenne et al., 2016, p. 1043)
  # -  $H_0$  states that both conditions share the same proportion
  X <- contingencyTableBF(
    table2x2,
    sampleType = "indepMulti",
    fixedMargin = "rows",
    priorConcentration = 1
  )

  # Unused draws, kept from the original script: they consume random numbers,
  # so removing them would change the posterior samples below (and thus the
  # reported Bayes factors)
  post_alt <- posterior(X, iterations = 10000)
  post_null <- posterior(1 / X, iterations = 10000)

  # BF_u0: evidence for  $H_u$  against  $H_0$  (computed analytically by BayesFactor;
  # X stores it on the log scale)
  bf_u0 <- exp(as.numeric(X@bayesFactor$bf))

  # Draw 10,000 posterior samples of the two proportions under  $H_u$ 
  # (these follow Beta(1 + yes, 1 + no) distributions in each condition)
  post.result <- BayesFactor::posterior(X, iterations = 10000)
  idx <- grep("omega", colnames(post.result))
  theta <- as.data.frame(post.result[, idx])
  # column 1:  $\theta_c$ , proportion of towel reuse in the control condition
  # column 2:  $\theta_e$ , proportion of towel reuse in the social-norm condition

  # Fit of  $H_1$  ( $\theta_e > \theta_c$ ) and of the complement  $H_c$  ( $\theta_e < \theta_c$ ):
  # the proportions of posterior samples in agreement with each hypothesis
  prop.consistent <- mean(theta[, 1] < theta[, 2])
  prop.complement <- mean(theta[, 1] > theta[, 2])

  # BF_1u = fit / complexity; the complexity is  $P(\theta_e > \theta_c)$  under the
  # uniform priors, which is 0.5 because the prior is symmetric

```

```

bf_1u <- prop.consistent / 0.5

# BF_10 by transitivity: BF_10 = BF_u0 * BF_1u
bf_10 <- bf_u0 * bf_1u

# BF_cu for the complement, with complexity 1 - 0.5 = 0.5;
# via BF_c0 = BF_u0 * BF_cu, so BF_cu = BF_c0 / BF_u0 = prop.complement / 0.5
bf_c0 <- bf_u0 * prop.complement / 0.5
bf_cu <- bf_c0 / bf_u0

# BF_1c by transitivity: BF_1c = BF_1u / BF_cu
bf_1c <- bf_1u / bf_cu

# Store the study-specific BFs
bf_1u_vec[i] <- bf_1u
bf_cu_vec[i] <- bf_cu
bf_10_vec[i] <- bf_10
bf_1c_vec[i] <- bf_1c
}

# Create data frame with results
results_df <- data.frame(
  Study = exList,
  bf_10 = bf_10_vec,
  bf_1u = bf_1u_vec,
  bf_1c = bf_1c_vec,
  bf_cu = bf_cu_vec
) %>%
  filter(Study != "Total")

# Create dataframe with only the relevant columns for the compute_joint_complements function
final_df <- results_df %>%
  select(Study, bf_1u, bf_cu) %>%
  mutate(P_theta_in_Hc = 0.5)

# T = nr of experiments
T <- nrow(final_df)

```

Study	$BF_{1,0}^{(t)}$	$BF_{1,u}^{(t)}$	$BF_{1,c}^{(t)}$	$BF_{c,u}^{(t)}$
Goldstein.2008.Ex1	1.453	1.941	32.670	0.059
Goldstein.2008.Ex2	2.023	1.974	75.336	0.026
Schulz2008.Ex2	0.354	1.720	6.148	0.280
Schulz2008.Ex3	0.580	1.858	13.045	0.142
Mair2010	0.316	1.225	1.579	0.775
Bohner2014.Ex1	0.065	0.667	0.501	1.333
Bohner2014.Ex2	0.064	0.036	0.018	1.964

Product Bayes factors

```
bf_10 <- prod(results_df$bf_10)
bf_1u <- prod(results_df$bf_1u)
bf_1c <- prod(results_df$bf_1c)
```

- $\text{BF}_{1,0}^{(\text{joint})} = 7.94e - 04.$
- $\text{BF}_{1,u}^{(\text{joint})} = 0.3581.$
- $\text{BF}_{1,c}^{(\text{joint})} = 2.85e + 03.$

BF against complete joint complement

1: Enumeration (Eq. 7)

```
set.seed(45)
res <- compute_joint_complements(
  final_df,
  replace_0_with = 0.001,
  round_res = FALSE
)

# Sanity check: BF_1u(joint) / BF_cu(joint) should equal the product of the BF_1c's
all.equal(res["BES_c", "bf_1u"] / res["BES_c", "bf_cu"], bf_1c)
```

```
[1] TRUE
```

```
bf_1c_star_enumeration <- res["BES_c", "bf_1u"] / res["BES_c_star", "bf_cu_star"]
```

2: Complexity weighting (Eq. 8)

- Check: with $c_1^{(t)} = 0.5$ for all studies, this should be close to equal weighting.
- Uses $\text{BF}_{1,u}^{(t)}$ and complexities $c_1^{(t)}$.

$$\text{BF}_{1,c^*}^{(\text{joint})} = \frac{\prod_{t=1}^T \text{BF}_{1,u}^{(t)} (1 - \prod_{t=1}^T c_1^{(t)})}{1 - \prod_{t=1}^T c_1^{(t)} \text{BF}_{1,u}^{(t)}}$$

```
c1 <- 1 - final_df$P_theta_in_Hc

bf_1c_star_formula_complexity <-
  prod(results_df$bf_1u) * (1 - prod(c1)) / (1 - prod(c1 * results_df$bf_1u))
```

3: Equal weighting (Eq. 9)

- Uses only $\text{BF}_{1,c}^{(t)}$.

$$\text{BF}_{1,c^*}^{(\text{joint})} = \frac{2^T - 1}{\prod_{t=1}^T (1 + 1/\text{BF}_{1,c}^{(t)}) - 1}$$

```
bf_1c_star_formula_equal <- (2^T - 1) / (prod(1 + 1 / results_df$bf_1c) - 1)
```

Summary

$\text{BF}_{1,0}^{(\text{joint})}$	7.942e-04
$\text{BF}_{1,u}^{(\text{joint})}$	0.3581
$\text{BF}_{1,c}^{(\text{joint})}$	2.846e+03
$\text{BF}_{1,c^*}^{(\text{joint})}$ (Eq. 7)	0.3563
$\text{BF}_{1,c^*}^{(\text{joint})}$ (Eq. 8)	0.3563
$\text{BF}_{1,c^*}^{(\text{joint})}$ (Eq. 9)	0.3563

Session info

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24)
```

```
Platform: aarch64-apple-darwin23
```

```
Running under: macOS Tahoe 26.5.2
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.12.0
```

```
locale:
```

```
[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
```

```
time zone: Europe/Berlin
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices datasets  utils      methods    base
```

```
other attached packages:
```

```
[1] stringr_1.6.0      tibble_3.3.1      here_1.0.2  
[4] dplyr_1.2.1        BayesFactor_0.9.12-4.8 Matrix_1.7-5  
[7] coda_0.19-4.1
```

```
loaded via a namespace (and not attached):
```

[1] jsonlite_2.0.0	compiler_4.6.1	renv_1.2.4	tidyselect_1.2.1
[5] Rcpp_1.1.2	MatrixModels_0.5-4	parallel_4.6.1	jquerylib_0.1.4
[9] yaml_2.3.12	fastmap_1.2.0	lattice_0.22-9	R6_2.6.1
[13] generics_0.1.4	knitr_1.52	rprojroot_2.1.1	bslib_0.12.0
[17] pillar_1.11.1	rlang_1.3.0	cachem_1.1.0	stringi_1.8.9
[21] xfun_0.61	sass_0.4.10	otel_0.2.0	cli_3.6.6
[25] withr_3.0.3	magrittr_2.0.5	digest_0.6.39	grid_4.6.1
[29] pbapply_1.7-5	mvtnorm_1.4-2	lifecycle_1.0.5	vctrs_0.7.3
[33] evaluate_1.0.5	glue_1.8.1	rmarkdown_2.32	tools_4.6.1
[37] pkgconfig_2.0.3	htmltools_0.5.9		